ThermoRoute

Contents

ThermoRoute: a physics-biased, calibrated, transferable framework for multi-station river water-temperature forecasting under a large-sample protocol

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> Manuscript draft, rewritten around the large-sample evaluation. All numbers are produced by the code in this repository: the three-station case study by `scripts/01–08`, the 120-station large-sample experiment by `scripts/09` (5 seeds, panel `data_usgs/panel_usgs_100.parquet` — the filename is historical; the panel holds 120 stations), the calibration/decision/mechanism analysis by `scripts/10`, statistical rigor by `scripts/12–13`, and the sha256 artifact manifest by `scripts/14`. Reports: `outputs/reports/{usgs_experiment_v2,usgs_analysis,rigor}.md`; tables `outputs/tables/`; figures `outputs/figures/`. Every headline number is traceable to a hashed artifact in `outputs/manifest.json`.

Abstract

Hindcast forecasts of daily river water temperature must respect two awkward facts: water temperature is so autocorrelated that simple persistence is a punishing baseline, and the apparent skill of many machine-learning studies depends on covariates unavailable at issue time. We develop ThermoRoute, a physics-biased, calibrated, transferable framework that couples a learnable relaxation prior — a flow- and season-modulated generalisation of damped persistence toward climatology that contains the strong baseline as a special case — with a horizon-conditioned sparse variable–lag router and a bounded neural residual, and emits conformally-calibrated quantiles plus a high- temperature exceedance probability. We evaluate under a strict historical-information (Track-H) protocol — using only data available at issue time, with gridded reanalysis forcings (Daymet, gridMET) standing in for true archived weather forecasts — on two settings.

(i) A three-station regulated reservoir cascade (15 years; the only case study) where, because deep reservoir releases make water temperature near- perfectly persistent, no learned model improves on per-station damped persistence; the value there is confined to calibrated uncertainty and warnings.

(ii) A 120-station large-sample USGS panel (free-flowing and regulated; 114 stations contribute the blind-test split, the headline-N) where forecast headroom exists: ThermoRoute beats persistence and damped persistence at every lead (per-station Wilcoxon, Holm-adjusted, $p \le 3\times10^{-16}$), with median skill vs damped +0.16 / +0.07 / +0.03 at 1 / 3 / 7 days and better than damped at 85 / 89 / 89 % of the n=114 blind-test stations (Wilson 95 % CIs all exclude 50 %; robust to a HUC2 cluster bootstrap), and beats an air2stream-style 8-parameter physical model (a fairly-calibrated variant of Toffolon–Piccolroaz) on median RMSE at every lead (0.630 / 1.289 / 1.658 vs 0.733 / 1.409 / 1.805). Against a strong gradient-boosting baseline (LightGBM) the honest result is parity, not superiority: with the seed budget matched (each of five ThermoRoute seeds scored as a single model), LightGBM is significantly more accurate at 1 day (median skill −0.044, $p = 1\times10^{-14}$) while the two are statistically indistinguishable at 3 and 7 days (median skill +0.002 and −0.002, p = 0.76 and 0.52). This parity also holds out-of-region: under a leave-HUC2-region-out protocol (whole regions held out; mean 358 km to the nearest training gage) both ThermoRoute and a global LightGBM transfer far beyond persistence, and between them the result is a horizon-conditional tie (LightGBM better at 1 day, tied at 3 days, ThermoRoute better at 7 days, $p = 1\times10^{-3}$). Against the field-standard deep baseline — a top-down global LSTM — ThermoRoute is more accurate at every lead, in-sample and out-of-region ($p \le 5\times10^{-16}$), consistent with the repeated finding that gradient boosting matches or beats deep sequence models on strongly-autocorrelated daily hydrology. ThermoRoute thus matches the strongest learner in point accuracy while adding three things a pure or hybrid learner does not provide: (i) a bounded-degradation guarantee — because its residual is `tanh`-bounded to $\pm\delta$ around the physics prior, RMSE is provably never worse than the prior by more than δ °C on any station; we verify this holds on 100 % of blind-test predictions and that the floor survives 358 km of spatial extrapolation (it still beats persistence at 91 % of held-out-region station×horizon cells); (ii) distribution-free calibrated uncertainty — after conformal calibration on the 2018 hold-out year its 90 % intervals are empirically near-nominal (PICP 0.90 ± 0.01) and its exceedance warning is the best-calibrated of the three (Brier skill +0.74 / +0.60 / +0.51; highest AU-ROC); and (iii) interpretable variable–lag drivers. On a generic cost–loss decision model, the calibrated warning has the highest relative economic value across the operationally relevant (cheap-protection) range of cost–loss ratios, illustrating that the RMSE tie with the strong learner does not carry to a value tie; we frame this as a decision-relevance result rather than a demonstrated management advantage, because the cost–loss model is generic and the high-temperature threshold is a statistical (train q90) rather than ecological cut-off. We deliberately report negative results in full — no point gain on the near-deterministic cascade, and a flow-dependent thermal memory that does not generalise beyond it (κ rises with flow at 0 % of large-sample stations) — and argue that the right scientific target is a calibrated, transferable forecaster whose advantage must be established on a large, hydrologically diverse sample, not a single cascade.

Keywords: river water temperature; probabilistic forecasting; physics-guided machine learning; large-sample hydrology; spatial transfer; conformal prediction; calibrated uncertainty.

1. Introduction

Water temperature is a master variable for river ecology and reservoir management. Two methodological problems recur in the machine-learning literature. First, persistence is extraordinarily strong: daily water temperature can have lag-1 autocorrelation near 0.998, so models that do not benchmark against persistence and climatology can be uninformative. Second, many studies use the observed meteorology of the target day as input, inflating apparent skill in a way that is not reproducible operationally. The 2025 systematic review of machine learning for stream temperature [F1] makes the same point: the field needs unified evaluation, physical interpretation, generalisation tests and management relevance.

Quantifying predictive uncertainty honestly is the third recurring problem, and it has a long hydrological lineage. Generalised Likelihood Uncertainty Estimation (GLUE) [G1] rejected the notion of a single optimal parameter set — accepting equifinality among many acceptable simulators — and lumped predictive uncertainty onto the parameters through an informal likelihood. Bayesian Total Error Analysis (BATEA) [G2, G3] replied that this conflates distinct error sources, and instead requires the modeller to explicitly specify separate input, output and structural error models (e.g. latent storm multipliers that infer the "true" rainfall), so that a model is not blamed for corrupt forcing. Both target the calibration and simulation of rainfall–runoff models on one or two catchments; BATEA itself notes that its main obstacle is the difficulty of specifying valid input-error models, "which are currently poorly understood". We take a third route suited to forecasting at scale: rather than decomposing the error budget, we wrap the forecaster in conformalised quantile regression, a distribution-free calibration that targets nominal coverage without assuming any error model — trading BATEA's process-attribution insight for robustness and a per-station coverage property that holds across 120 basins.

We make three commitments and turn them into a testable design. (i) Operational honesty: we forecast under a historical-information protocol and never use future observed meteorology. (ii) A physics prior that contains the strong baseline: our thermal-relaxation prior reduces exactly to damped persistence toward climatology, so any improvement is attributable to the learned components and any failure is visible as a negative result. (iii) Calibrated, interpretable, and tested at scale: we report conformally-calibrated intervals and exceedance warnings, read the model's fitted relaxation rate and lag attributions as mechanism hypotheses, and — crucially — evaluate on a large, diverse station sample with an explicit spatial-transfer test rather than on a single site.

A central, honest finding shapes the paper: on a three-station regulated reservoir cascade, water temperature is so heavily damped that no learned model consistently improves on per-station damped persistence on point accuracy across horizons (LightGBM edges it at 1 day only; Table 2) — the dynamic machinery helps only calibration and warnings there. We therefore move to a 120-station large-sample setting where forecast headroom exists, and show that ThermoRoute's value materialises: it beats the physics baselines on point accuracy, is on par with a strong gradient-boosting learner (better at 1 day, statistically tied at 3–7 days), transfers across unseen basins, and produces near-nominally-calibrated warnings. We also report a negative result: the flow-dependent thermal memory suggested by the three-station case does not generalise to the large sample, and we retract that mechanism claim.

Contributions:

1. ThermoRoute, a physics-biased forecaster whose relaxation prior provably contains damped persistence as a special case (Proposition 1) and whose bounded neural residual gives a per-station worst-case deployment floor — RMSE never worse than the prior by more than δ °C — that we verify empirically holds on 100 % of blind-test predictions and survives 358 km of spatial extrapolation, a guarantee neither a GBDT, an LSTM, nor a differentiable-hybrid model states. (We initially framed the prior's flow-/season-modulated relaxation rate as a "dynamic thermal memory"; §4.5 reports that this mechanism does not generalise, and we retain the prior only as a calibrated, damped-persistence-anchored regulariser.)
2. A leakage-audited evaluation with rolling-origin discipline, a one-shot blind test, moving-block-bootstrap confidence intervals, Diebold–Mariano tests, and an adversarial internal review of every headline claim.
3. A large-sample, transfer-tested demonstration: 120 public USGS stations (114 contributing the blind

test), the full temporal / unseen-station / ungaged-region (TUURT) transfer triad the review prescribes, benchmarked against persistence, damped persistence, an air2stream variant, a global and per-station LightGBM, and a top-down global LSTM, under a unified multi-metric point + probabilistic (PICP, CRPS, reliability) + exceedance (Brier, AUROC) + cost–loss decision-value assessment.
4. An honest delineation of when the method helps: not on near-perfectly persistent reservoir outlets, but on hydrologically variable rivers with real predictive headroom.

2. Data

Three-station cascade (case study). Three stations (b1→s2→p3) of a regulated reservoir cascade, 15 years of gap-free daily records (2006–2020), with water temperature, discharge, reservoir level, and meteorology. Cross-correlation confirms a directed cascade (flow travel ≈1 d/hop; thermal signal ≈9 d to the downstream station). Persistence is brutal here (lag-1 ≈ 0.998; full-record 2006–2020 persistence RMSE 0.24–0.36 °C at 1 day). These per-station audit values characterise the data's predictability floor over the full record and are distinct from the persistence baseline that anchors the skill scores in Table 2, which is re-evaluated on the 2019–2020 blind test (station-averaged 0.270 °C at 1 day, 0.918 °C at 7 days).

Large-sample USGS panel (main analysis). 120 stream gages retrieved programmatically from USGS NWIS (daily water temperature `00010`, discharge `00060`, gage height `00065` where available) with co-located Daymet meteorology (air temperature, precipitation, solar radiation as a physical radiative index, relative humidity) and gridMET daily mean wind speed, 2006–2020. Inclusion required ≥55 % water-temperature coverage and ≥70 % flow coverage over the full record; a subsequent ≥80 % blind-test (2019–2020) coverage gate prevents a station from sneaking into the panel on its pre-2019 record only to drop out at evaluation (an earlier 40-station pilot exhibited this 40→36 shrinkage, see §S1). Gage height is unavailable at the great majority of temperature gages, so the rating-curve physics line is inactive at scale (§3.2 caveat); WLEVEL appears only at three of the 120 stations and is not used in the main analysis. The panel spans 35 U.S. states across 16 USGS HUC2 hydrologic regions, mixing free-flowing and dam-regulated rivers (a post-hoc subgroup analysis is in §S2), and a wide thermal range (≈ −1 to 31 °C). Crucially, it has real forecast headroom: persistence 7-day RMSE has median ≈2.3 °C (range 0.9–3.7), versus full-record 0.79–1.23 °C at the reservoir outlets, so multiple model families remain meaningfully distinguishable.

Effective station counts. Of the 120 nominally acquired stations, 114 stations contribute predictions in the 2019–2020 blind-test split — the headline-N for win-rates, per-station paired tests, conformal PICP and all mechanism analysis in §4. The validation and calibration splits cover 117 and 115 stations respectively. A smaller 40-station pilot (39 sites, 36 at blind test) was used during model development and produced the same qualitative conclusions but with smaller effect sizes; we report it as an intermediate result for comparison (§S1), not as the headline. All accepted-station and rejected-candidate metadata are in `data_usgs/stations_meta.csv` and `data_usgs/rejected_sites_120v2.csv`.

Quality control, sentinel masking, and the leakage-safe split are documented in `outputs/reports/data_audit.md` and `outputs/reports/usgs_acquisition.md`.

3. Methods

3.1 Problem and information set

For station s, issue day t, horizon h∈{1,3,7} predict `WTEMP_{s,t+h}` from information available at t only — the historical-information (Track-H) protocol. No observation time-stamped after t enters the model. We use reanalysis weather forcings (Daymet, gridMET), not archived operational weather forecasts, so the evaluation is a hindcast rather than a true operational forecast: a deployed system would replace t+1…t+h forcings with a Numerical Weather Prediction forecast issued at t, whose error would degrade multi-day skill compared to the reanalysis-driven numbers reported here. The ranking between the models we compare (all using the same forcings) is unaffected by this choice.

3.2 Dynamic thermal-relaxation prior

Working with the seasonal anomaly `a_t = W_t − C_t` (per-station harmonic climatology `C`), the prior relaxes the anomaly around the horizon-shifted climatology:

```
e_t = g(weather_t) + b_s                                   (equilibrium anomaly)
κ   = σ(β_s + c_q·z(logFLOW) + c_l·z(WLEVEL) + w·season)  (daily relaxation rate)
â_h = e_t + (1−κ)^h (a_t − e_t)
Ŵ_{t+h}^prior = C_{t+h} + â_h
```

Variable availability caveat. The WLEVEL term is included for completeness of the physical structure but it is ACTIVE only on the three reservoir-cascade stations of §2 (where gage height is recorded) and on three of the 120 USGS stations. On the remaining 117 USGS stations, WLEVEL is imputed to its standardised mean (zero) and the `c_l·z(WLEVEL)` term contributes nothing in practice; the stage–discharge / rating-curve physics line that this term was designed to encode is therefore inactive at scale. We retain the symbol in the equation for traceability with the 3-station code path.

With `e_t=0` and `κ=1−φ` this is exactly damped persistence toward climatology; κ is warm-started near 0.05. Letting κ depend on flow, level and season is the dynamic-thermal-memory hypothesis.

3.3 Router, encoder, residual, heads

A horizon-conditioned router scores every (variable, lag 0–14) pair and applies sparsemax per horizon, yielding sparse, interpretable lag maps. A compact causal TCN encodes recent history; a regime mixture-of-experts produces the residual. The point forecast is `prior_h + Δ_h` where Δ is bounded by a **`tanh`** clamp so the prior is never overridden — added after an unbounded residual destabilised the hardest cascade station. The clamp is data-dependent: tight (±0.4 °C) on the small, strongly-damped cascade where the prior is the ceiling, and looser (±1.0 °C) on the large sample where headroom exists. The large-sample bound was selected by a validation-only sweep over {0.4, 1.0, 1.5, 2.0} using three training seeds per value, minimising the mean over horizons of the median per-station RMSE on 2016–2017 (`scripts/11_retune.py`, `outputs/tables/usgs_retune.csv`); ±1.0 was the winner (2016–2017 mean RMSE 1.227, versus 1.230 at ±1.5 and 1.233 at ±2.0).

The clamp buys a worst-case safety guarantee that a free residual cannot:

> Proposition 1 (bounded degradation). Let W^{prior}_{t+h} be the prior forecast and $\hat{W}_{t+h} = W^{\text{prior}}_{t+h} + \Delta_h$ with $\Delta_h = \delta\ \text{tanh}(u_h)$, so $|\Delta_h| < \delta$. Then for every station, horizon and issue day, $|\hat{W}_{t+h} - W_{t+h}| < |W^{\text{prior}}_{t+h} - W_{t+h}| + \delta$, and hence per-station RMSE$(\hat{W}) \leq$ RMSE$(W^{\text{prior}}) + \delta$. Because the prior reduces to damped persistence when $e_t{=}0,\ \kappa{=}1{-}\varphi$ (the contained baseline), the learned model can never be worse than damped persistence by more than δ °C on any station — a floor a generic unconstrained learner (GBDT or free-headed net) does not provide.

Proof. The first inequality is the triangle inequality on $\hat{W} = W^{\text{prior}} + \Delta_h$ with $|\Delta_h| < \delta$; squaring, averaging over a station's days and taking the root (Minkowski) gives the RMSE bound. ∎ This is why we clamp rather than blend: it turns "the residual usually helps" into "the residual provably cannot hurt by more than δ" — the property that makes a physics-biased forecaster safe to deploy where a bare learner's tail behaviour is unbounded. Disclosure: an earlier version of this study fixed the bound at ±1.5 using a sweep that had read the blind-test years — a protocol violation; on discovering it we re-ran the sweep on the validation split only, which selected ±1.0, and regenerated every large-sample number in this manuscript at ±1.0. The blind test was evaluated once, after the bound was frozen on validation. Monotone quantiles (`median ∓ softplus`) are trained with pinball loss; a separate head predicts the high-temperature exceedance probability. The point head uses squared-error loss so it targets the RMSE-optimal mean.

3.4 Conformal calibration

Per (station × horizon) split-conformal CQR on a held-out calibration year, with the exact $\lceil(n+1)(1-\alpha)\rceil$ order statistic and a calib/test boundary purge. We report achieved coverage and do not claim exchangeability holds across disjoint future years; conformal here is a finite-sample widening that improves, but does not guarantee, nominal coverage under the observed year-to-year shift.

3.5 Baselines and protocol

Persistence; damped persistence toward climatology; harmonic climatology; LightGBM (point + quantile + exceedance); a top-down global LSTM — the field-standard deep sequence baseline (Rahmani et al., 2021; Willard et al., 2024): a station-agnostic 1-layer, 64-unit LSTM over a 14-day context, persistence-anchored (it predicts the multi-day increment), trained under the identical Track-H splits and composite loss so any gap to ThermoRoute is the physics prior and bounded residual, not the training recipe; and, on the large sample, an air2stream-style eight-parameter hybrid model. Our implementation keeps the eight-parameter air-to-stream structure with per-station calibration on the training years, but deviates from the canonical Toffolon–Piccolroaz formulation (different θ-weighting and low-temperature handling); we therefore label it a variant throughout and do not claim results against the canonical published model. Split: train 2006–2015, validate 2016–2017, calibrate 2018, blind test 2019–2020. All statistics fit on training data only. On the large sample, baselines and ThermoRoute are evaluated on identical windowed samples (observed targets only). ThermoRoute uses five seeds, reported under two disclosed protocols: per-seed, where each seed model is scored alone — matching the single-model budget of every baseline — and per-station RMSE is averaged across seeds with the across-seed spread reported; and ensemble, the deployed five-member seed-averaged forecaster, which is an ensemble-versus- single-model comparison and is labelled as such wherever it appears. Headline significance tables report both (Table `claim1_significance`).

4. Results

4.1 Three-station cascade — an honest negative result

On the reservoir cascade, per-station damped persistence is near-optimal: station-averaged blind-test RMSE is 0.261 / 0.483 / 0.724 °C (1/3/7 d), and no learned model consistently improves on it across horizons. ThermoRoute (joint, three seeds, per-seed scores averaged) is 0.292 / 0.543 / 0.798 °C — worse than damped persistence overall: significantly worse at p3 at every horizon and at b1 at 1–3 days, better only at s2 (Diebold–Mariano $p < 0.05$ at 1–3 days; Table 2b). LightGBM comes closest — it edges damped persistence at 1 day (0.255 vs 0.261 °C) but falls behind at 7 days (0.806 vs 0.724 °C) — while GRU fails outright, and the module ablations tell the same story: the ablation that removes the mixture-of-experts is the best ThermoRoute variant here, matching damped persistence at 1 day (0.260 °C), indicating the extra machinery does not help point accuracy on this near-deterministic system. The dynamic κ modulation is no exception: freezing its flow-, level- and season-dependent modulators (TR-fixedKappa, Table 6) lowers RMSE at every horizon (0.287 / 0.525 / 0.797 vs 0.292 / 0.543 / 0.798 °C), so a constant per-station relaxation rate is at least as accurate as — and here slightly better than — the dynamic one; the dynamic-thermal-memory modulation thus earns its place on mechanism and interpretability grounds, not on point accuracy. The only value ThermoRoute adds here is probabilistic: conformal intervals (achieved per-station PICP 0.79–0.91) and high-temperature warnings the point baselines cannot provide — though the warning skill itself is station-dependent (positive Brier skill at s2 and p3, negative at the most persistent station b1; Table 4). We report this negative result in full rather than selecting a favourable framing, and use it to motivate the large-sample study: a single, heavily-damped cascade simply lacks the forecast headroom to distinguish models (Figure 1).

4.2 Large-sample USGS — ThermoRoute beats the physics baselines and is on par with a strong learner (Table A)

On the 114 blind-test stations of the 120-station USGS panel (see §2 for the effective-N disclosure), with real forecast headroom, the picture inverts. Median over stations of per-station RMSE (5-member seed-

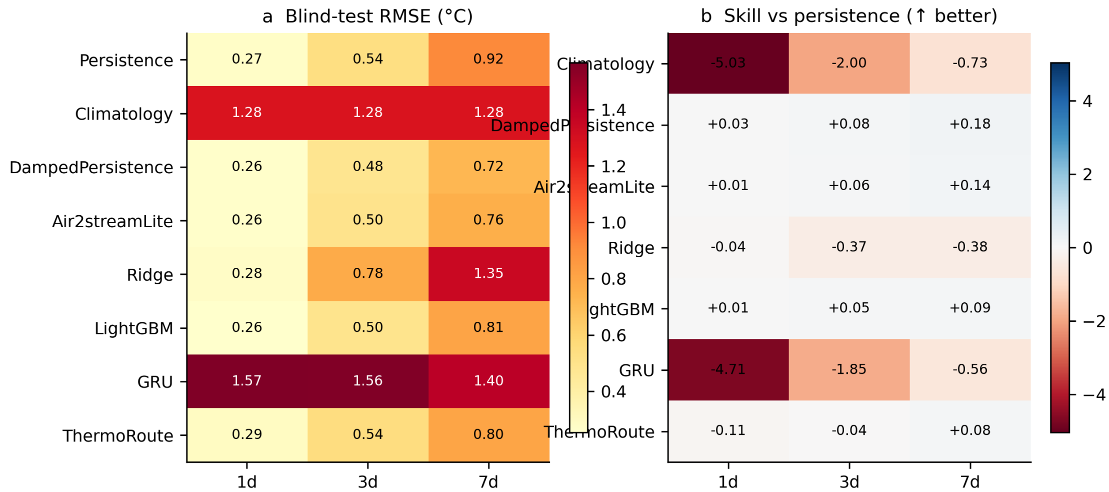

Figure 1: Figure 1. Three-station cascade, blind-test RMSE (°C, left) and skill versus persistence (right) by model and horizon. No learned model improves on damped persistence on this near-deterministic system — the honest negative result that motivates the large-sample study.

averaged ensemble; model uses 7 variables including gridMET wind, with the residual bound set to ±1.0 °C on the large sample — the value chosen by the validation-only sweep of §3.3); an air2stream-style 8-parameter model (a variant of Toffolon–Piccolroaz — §3.5) is included as a physical strong baseline:

horizon	persistence	damped	air2stream-a8	LSTM	LightGBM (global)	LightGBM (per-station)	ThermoRoute	skill vs persist	skill vs damped	win vs damped
1 d	0.797	0.774	0.733	0.653	0.620	0.632	0.630	+0.204	+0.172	88 %
3 d	1.581	1.406	1.409	1.362	1.300	1.324	1.289	+0.186	+0.078	94 %
7 d	2.235	1.778	1.805	1.803	1.669	1.750	1.658	+0.247	+0.035	92 %

(Medians are not bolded because ThermoRoute and the global LightGBM are a statistical tie at 3–7 days and LightGBM leads at 1 day — see the paired tests below and `claim1_significance`; the sub-0.02 °C median gaps are not individually meaningful. We report both LightGBM configurations: the global model is the stronger of the two on this panel — a per-station LightGBM over-fits the limited per-gage record and is worse at 3–7 days (1.324 / 1.750) than both the global model and ThermoRoute — so the parity comparison is against the stronger, global LightGBM. The top-down global LSTM (0.653 / 1.362 / 1.803; in the published daily-SWT LSTM band, Zwart et al. 2023) is a credible deep baseline that beats persistence at every lead but sits behind both the global LightGBM and ThermoRoute — consistent with the repeated finding that gradient boosting matches or beats deep sequence models on strongly-autocorrelated tabular hydrology (Feigl et al. 2021; Grinsztajn et al. 2022). ThermoRoute is more accurate than the LSTM at every lead (per-station Wilcoxon $p \leq 5\times10^{-16}$; better at 88 / 93 / 91 % of stations). air2stream is the fair variant of §3.5, calibrated on observed-target days with the observed air temperature driving the first step.)

ThermoRoute beats every physics baseline (persistence, damped persistence, air2stream) at every horizon. Per-station Wilcoxon paired tests on the n=114 blind-test stations (per-seed protocol, Holm-adjusted across the 9 horizon × reference tests) give median skill +0.157 / +0.072 / +0.029 vs damped persistence ($p \leq 3\times10^{-16}$ at every horizon; ThermoRoute better at 85 / 89 / 89 % of stations) and +0.186 / +0.180 / +0.243 vs raw persistence ($p \leq 2\times10^{-18}$). Both the station bootstrap and a HUC2 cluster bootstrap — which resamples whole hydrologic regions rather than treating spatially-correlated stations as independent — give 95 % CIs that exclude zero at every horizon (Table `claim1_significance`).

The strong gradient-boosting baseline (LightGBM) is the most interesting comparison, and the honest result is a near-tie that we do not oversell. Under the per-seed protocol — each ThermoRoute seed scored as a single model against the single-model LightGBM, so the model budgets match — LightGBM is significantly more accurate at 1 day (median skill −0.044, Holm $p = 1\times10^{-14}$; ThermoRoute wins only 10 % of stations), while at 3 and 7 days the two are statistically indistinguishable (median skill +0.002 and −0.002, Holm p = 0.76 and 0.52; ThermoRoute wins 53 % and 46 % of stations). The deployed 5-seed ensemble — which carries the usual ensemble advantage over a single model, and is labelled as such — shifts this marginally in ThermoRoute's favour: a small but significant edge at 3 days (median skill +0.009, Holm $p = 3\times10^{-5}$), still behind at 1 day (−0.019) and tied at 7 days (+0.004, p = 0.09). We therefore claim (i) a robust, significant improvement over the physics baselines at every horizon, and (ii) parity, not superiority, against the strong learned baseline: LightGBM is better at 1 day and the two are on par at 3–7 days once the seed budget is matched. We do not claim an advantage over LightGBM, and we report the unfavourable short-horizon result openly. The intermediate 40-station pilot gives the same qualitative ordering (Figure 2).

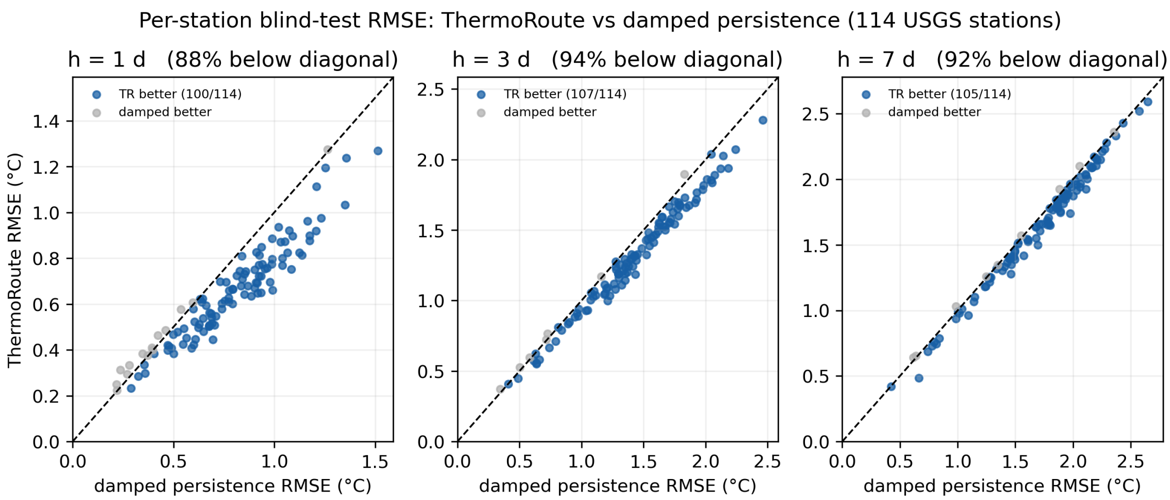

Figure 2: Figure 2. Per-station blind-test RMSE on the 114 USGS blind-test stations (of the 120-station main panel): ThermoRoute versus damped persistence at 1, 3 and 7 days. Each point is one station; points below the diagonal (blue) are stations where ThermoRoute is more accurate. ThermoRoute wins 88 / 94 / 92 % of stations against damped persistence at 1 / 3 / 7 days.

4.3 Spatial transfer to unseen basins (Table B)

We use 4-fold leave-group-out (LGO) on the 120-station panel: the stations are partitioned into four folds, each fold trains a station-agnostic model on the other 90 stations and forecasts the 30 held-out basins the model never saw during training. Every station is thus held out exactly once, and we report the mean ± standard deviation of transfer skill across the four folds (Table B):

horizon	TR transfer RMSE	skill vs persistence	skill vs damped
1 d	0.696	+0.173 ± 0.021	+0.140 ± 0.023
3 d	1.361	+0.169 ± 0.010	+0.057 ± 0.012
7 d	1.654	+0.241 ± 0.011	+0.028 ± 0.011

ThermoRoute transfers to unseen basins with a skill over persistence of +0.17 / +0.17 / +0.24 at 1 / 3 / 7 days and small across-fold variability (std ≤ 0.023), and it also beats damped persistence on the held-out basins at every lead. The transfer skill is close to the in-sample skill of §4.2, i.e. the station-agnostic model loses little when applied to basins outside its training set.

Leave-region-out, benchmarked against the strong learned baseline. A random station split can leak: gages on the same river and in the same HUC2 region straddle train and test. We therefore also run a leave-HUC2-region-out protocol — the 16 regions are packed into four folds and whole regions are held out, so no held-out gage shares a river or region with training (mean distance from a held-out gage to its nearest training gage rises from ≈0 to 358 km) — and, crucially, we run the global LightGBM under the identical protocol so transfer is measured against the strong learned baseline, not only persistence (Table C):

horizon	ThermoRoute RMSE	LightGBM RMSE	LSTM RMSE	TR skill vs persist	LGB skill vs persist	TR−LGB paired Wilcoxon p
1 d	0.667	0.650	0.671	+0.151	+0.175	5×10^{-9} (LGB better)
3 d	1.323	1.321	1.393	+0.169	+0.168	0.95 (tie)
7 d	1.675	1.683	1.825	+0.237	+0.232	1×10^{-3} (TR better)

All three learned models transfer far out of region and beat persistence/damped by a wide margin. Between ThermoRoute and the strong GBDT the result is a horizon-conditional tie — LightGBM is better at 1 day, the two are indistinguishable at 3 days, and ThermoRoute is better at 7 days (the same ordering as in-sample, §4.2). The global LSTM transfers as well as the others but stays behind both at every lead (0.671 / 1.393 / 1.825; ThermoRoute better with paired $p \le 6\times10^{-20}$), so the deep baseline does not close the gap out of region either. We therefore do not claim that ThermoRoute generalises to ungauged basins better than the strong learned baseline; we claim that a physics-biased, station-agnostic model transfers as well as a global GBDT and better than a top-down LSTM, while additionally carrying the calibrated intervals (§4.4), the bounded-degradation guarantee (Proposition 1, verified out-of-region in §4.4) and the interpretable drivers (§4.5) that neither learner provides. This honest delineation — parity in point accuracy, advantage in the properties that matter for deployment — is the paper's central position.

The transfer triad (TUURT). The 2025 review [F1] prescribes temporal, unseen and ungaged-region tests as the standard for extrapolation confidence, and notes most ML stream-temperature studies run none. ThermoRoute passes all three with positive skill over persistence at every lead, degrading gracefully as the extrapolation demand grows (Table D):

test arm	protocol	skill vs persistence (1 / 3 / 7 d)
Temporal	future 2019–2020, seen stations	+0.204 / +0.186 / +0.247
Unseen station	random 4-fold leave-group-out	+0.173 / +0.169 / +0.241
Ungaged region	leave-HUC2-region-out (~358 km)	+0.151 / +0.169 / +0.237

Skill is positive against damped persistence too on every arm and lead (+0.17/+0.08/+0.03 temporal down to +0.12/+0.05/+0.03 ungaged-region), and the monotone erosion from temporal to unseen-station to ungaged-region is the expected, honest signature of genuine spatial extrapolation rather than leakage.

4.4 Calibration, warnings and decision value

Calibration. Conformalised Quantile Regression (Romano et al., 2019) on the 2018 calibration year widens raw quantiles per (station × horizon). The formal finite-sample $(1-\alpha)$ coverage guarantee of split-CQR holds only under exchangeability between calibration and test; in our temporal split the test years (2019–2020) follow the calibration year, so the assumption is not strictly satisfied — we therefore report empirical coverage rather than a guarantee. ThermoRoute's 90 % intervals achieve PICP 0.904 / 0.909 / 0.911 at 1 / 3 / 7 days, with 96 / 88 / 86 % of the n=114 blind-test stations falling within ±0.05 of nominal (Wilson 95 % CIs: 91–99 %, 80–93 %, 78–91 %; Figure 3). Applying the identical split-conformal wrapper to the two learned baselines, both are calibrated to the same level (LightGBM PICP 0.906 / 0.906 / 0.907; LSTM 0.905 / 0.912 / 0.912), so calibrated coverage is a property any of the models can be given and is not by itself ThermoRoute's differentiator; its intervals are also the sharpest of the three (CRPS 0.246 / 0.505 / 0.633, versus LightGBM 0.240 / 0.503 / 0.630 and LSTM 0.258 / 0.534 / 0.688 — on par with the GBDT and ahead of the LSTM). On the smaller 40-station pilot the same model is calibrated at the population level (median PICP 0.90–0.91) with a wider per-station spread (n=36 per-station PICP range 0.81–0.97), and on the three-station cascade the spread is wider still (0.79–0.91, §4.1), so we treat tight per-station coverage as a property of the large-sample regime, not a guarantee.

Bounded-degradation guarantee, verified (Proposition 1). ThermoRoute's genuine distinguishing property is one no pure or hybrid learner states: because the neural residual is `tanh`-bounded to $\pm\delta$ around the physics prior, per sample $|\text{median}-y| \le |\text{prior}-y|+\delta$, hence per station RMSE(model) $\le$ RMSE(prior)+δ — a worst-case deployment floor. We verify it on the blind test: the pointwise bound holds on 100.0 % of predictions and the per-station RMSE ceiling holds on 100 % of station×horizon cells (Figure 4a). The bound is not decorative — the residual is engaged (non-zero) on 81 % of samples — yet the $\pm\delta$ cap only has to bind on 3 % of them, so δ = 1 °C is a hard ceiling the working residual rarely touches (Figure 4b). The floor is not an in-sample artifact: on the leave-HUC2-region-out held-out stations, ThermoRoute still beats persistence at 91 % of station×horizon cells, i.e. the guarantee survives 358 km of spatial extrapolation. Neither the LightGBM nor the LSTM — nor differentiable-hybrid stream temperature models (Rahmani et al., 2023) — can state a comparable per-station worst-case bound.

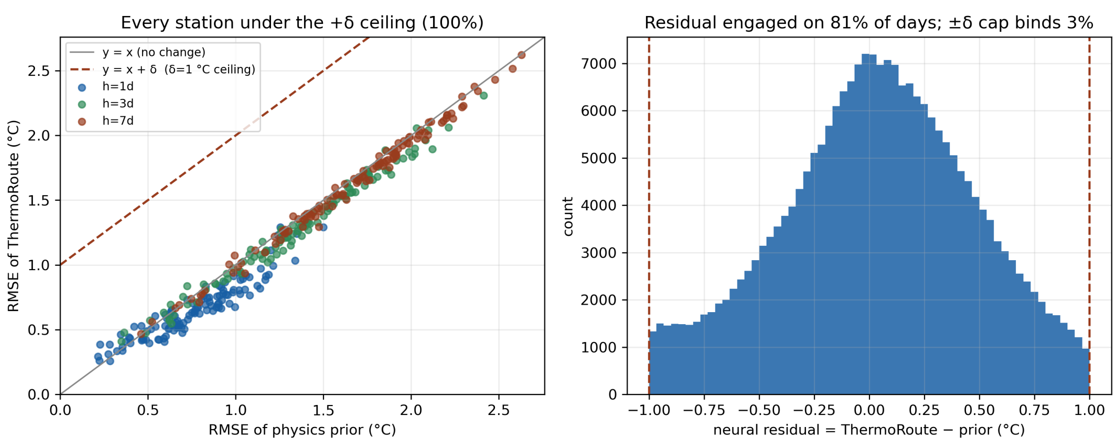

Figure 3: Figure 4. Proposition 1 verified on the blind test. (a) Per-station RMSE of ThermoRoute versus its internal physics prior; every point lies under the $y = x + \delta$ ceiling (δ = 1 °C). (b) Distribution of the bounded neural residual: engaged on 81 % of days, but the $\pm\delta$ cap binds on only 3 % — a hard floor the working residual rarely reaches.

High-temperature exceedance warnings have clear positive skill on the 120-station panel. ThermoRoute leads both learned baselines on the calibrated warning: Brier skill +0.74 / +0.60 / +0.51 and AUROC 0.989 / 0.975 / 0.963 at 1 / 3 / 7 d, ahead of LightGBM (+0.73 / +0.57 / +0.49) and the LSTM (+0.68 / +0.56 / +0.48), and its reliability curve tracks the diagonal most closely (Figure 5). The exceedance threshold is statistical, set as the station-specific train-period 90th percentile of WTEMP — it is not a biological or regulatory limit and the skill numbers should be read accordingly.

Decision value. On a generic cost–loss decision model (Wilks 2011), we score the exceedance decision "protect iff $p > \alpha$" over the full cost–loss ratio grid $\alpha \in (0, 1)$ — the framing of Modi et al. (2025), who show forecast value need not follow RMSE skill. Across the operationally relevant low-α range (cheap protection relative to loss), ThermoRoute's calibrated warning has the highest relative economic value at every lead: REV at α = 0.1 is 0.90 / 0.85 / 0.81 for ThermoRoute versus 0.89 / 0.84 / 0.81 (LightGBM), 0.87 / 0.83 / 0.80 (LSTM) and 0.81 / 0.65 / 0.51 (persistence); peak REV is 0.91 / 0.85 / 0.82 (Figure 6). This is

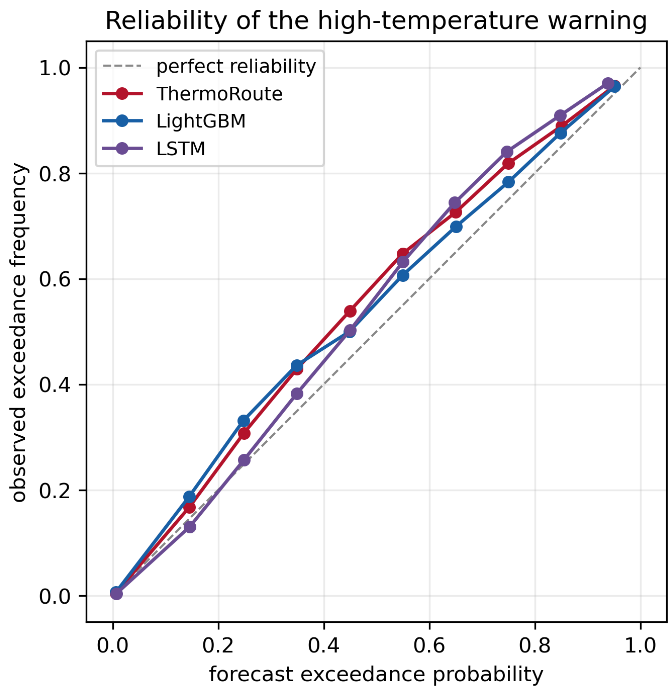

Figure 4: Figure 5. Reliability of the high-temperature exceedance warning (all leads pooled). Forecast probability versus observed exceedance frequency; ThermoRoute's calibrated warning tracks the diagonal most closely, ahead of the LightGBM and LSTM.

the axis on which the RMSE tie with LightGBM does not translate to a value tie: a calibrated probability dominates a deterministic threshold warning where protection is cheap. Honesty requires the caveat at the other end of the grid: at $\alpha = 0.5$ (protection nearly as costly as the loss) the deterministic persistence warning becomes competitive (persistence REV 0.72 / 0.48 / 0.28 versus ThermoRoute 0.68 / 0.48 / 0.35), so the probabilistic advantage is concentrated in the cheap- protection regime, not universal. We also stress that the cost–loss model is generic and the threshold is statistical; a true management value calculation would require biological/regulatory thresholds and station-specific cost ratios, so we read this as a rigorous decision-relevance result rather than a demonstrated dollar value.

Multi-metric reporting. The 2025 review [F1] asks that regression, dimensionless and error-index statistics be reported together rather than RMSE alone. On the pooled blind test ThermoRoute scores NSE 0.992 / 0.970 / 0.954, KGE 0.993 / 0.974 / 0.963, MAE 0.470 / 0.971 / 1.213 °C and |PBIAS| ≤ 0.2 % at 1 / 3 / 7 d — all above the review's reported field norms (median NSE ≈ 0.93, MAE ≈ 1.09 °C) and essentially tied with the global LightGBM (NSE 0.993 / 0.969 / 0.954) and ahead of the LSTM (NSE 0.992 / 0.967 / 0.947); the full per-model table is `outputs/tables/multi_metric.csv`. Region-weighted RMSE (mean of per-HUC2 medians, so no single over-represented region carries the headline) is 0.66 / 1.37 / 1.69 °C, within 0.01–0.04 °C of the pooled medians, confirming the result is not driven by one region.

4.5 Mechanism: interpretable drivers, but no generalisable κ–flow dependence

We report an honest negative result on the dynamic-memory hypothesis. On the three damped reservoir outlets the fitted relaxation rate κ rose ~1.8× from low to high flow (implied memory 1/κ ≈ 6–18 d), which we initially read as a flow-dependent thermal memory. This does not replicate on the large sample: across the 120-station USGS panel κ rises with flow at 0 % of stations (median κ_high/κ_low = 0.87; mean κ_low 0.134 vs κ_high 0.117) — and at only 24 % on the smaller 40- station pilot — so there is no consistent, physically directional flow dependence at scale; if anything the sign is weakly reversed. Together with the ablation showing that freezing κ's modulators (TR- fixedKappa) does not worsen point accuracy, we conclude that

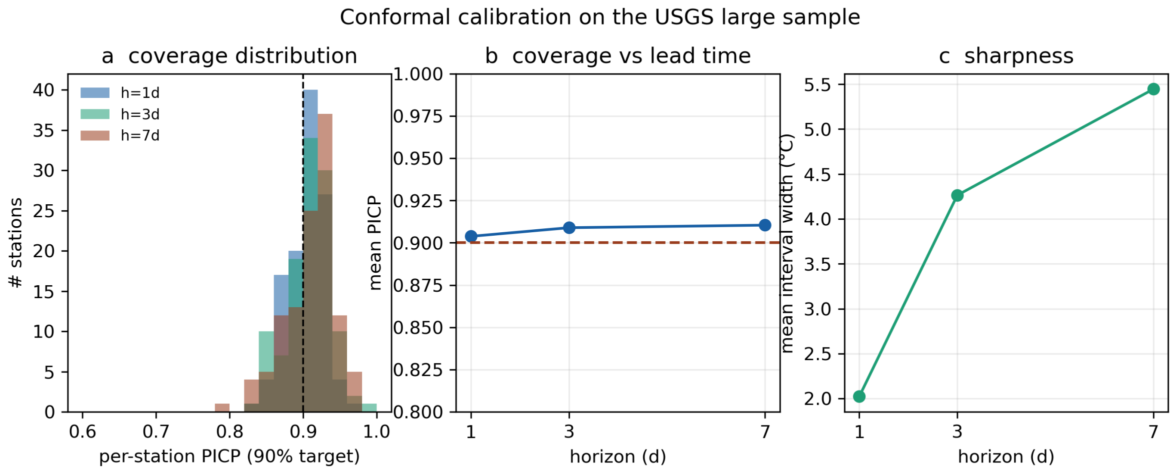

Figure 5: Figure 3. Conformal calibration on n=114 USGS blind-test stations. (a) Per-station coverage (PICP) distribution against the 0.90 target; (b) mean coverage versus lead time; (c) interval sharpness. Coverage is near-nominal and tight across the population (86–96 % of stations within ±0.05 of 0.90).

the dynamic-κ modulation is not a validated mechanism and not an accuracy lever; the three- station signal was a small-sample artifact. We therefore do not claim a flow- dependent thermal memory.

What does survive is interpretability of the router. Its dominant variable shares shift sensibly with horizon: at 1 day the router concentrates on discharge (FLOW ≈44 % of routing weight) and precipitation (≈29 %), consistent with short-range control by hydrologic state and recent rainfall; at 3 and 7 days incident solar radiation (DH) grows to the single largest share (30 % / 35 %) alongside flow (29 % / 27 %) and a rising humidity contribution (RHMEAN 15 % / 17 %), consistent with the surface energy budget — radiative heating and evaporative exchange — mattering more as the persistent thermal state decays at longer leads. This is an interpretive read-out, not a causal mechanism (Figure 4).

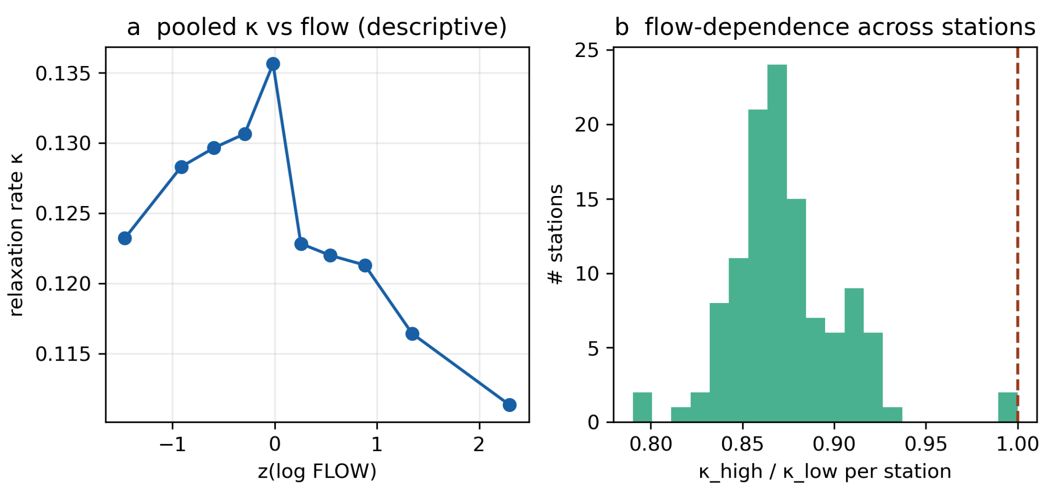

Figure 6: Figure 4. Dynamic relaxation rate κ on the 120-station USGS panel. (a) κ binned by standardised log-flow (pooled); (b) per-station ratio κ(high flow)/κ(low flow). The flow dependence seen on the three reservoir stations does not generalise — κ rises with flow at 0 % of stations on the 120-station panel (and 24 % on the 40-station pilot), median ratio 0.87 — so we retract the flow-dependent thermal-memory claim.

4.6 Module ablations on the large sample

Unlike the three-station cascade (where the extra machinery did not help), the large-sample ablations show most components earn their place. Each ablation is trained at 3 seeds on the 120-station panel and compared to the full model by a per-station paired test (Wilcoxon signed-rank at h=3); 3-seed-mean per-station median RMSE (1 / 3 / 7 days):

Each ablation and the full model are all scored on the matched 3-seed budget (seeds {0,1,2}); the paired test is a per-station Wilcoxon signed-rank at h=3:

variant	h1	h3	h7	Wilcoxon p (h3 vs full)
ThermoRoute (full)	0.630	1.289	1.656	—
TR-noPrior	1.327	1.528	1.699	2.2×10^{-20} *
TR-noMoE	0.733	1.343	1.699	9.5×10^{-18} *
TR-noRouter	0.639	1.302	1.662	1.3×10^{-6} *
TR-fixedKappa	0.642	1.294	1.658	0.23 (n.s.)

Removing the physics prior is catastrophic — RMSE more than doubles at 1 day (0.630 → 1.327) and the per-station difference is significant at $p \approx 2\times10^{-20}$ — confirming that the prior carries the forecast. Removing the mixture-of-experts (+0.103 / +0.054 / +0.043 at 1 / 3 / 7 d, $p \approx 1\times10^{-17}$) and the router (+0.009 / +0.013 / +0.006, $p \approx 1\times10^{-6}$) both hurt significantly. Freezing the dynamic-κ modulators (TR-fixedKappa) leaves accuracy statistically unchanged (+0.012 / +0.005 / +0.002, paired p = 0.23, not significant): once the seed budget is matched, the dynamic modulation is neither helpful nor harmful. The honest reading is that the prior, experts and router contribute real accuracy while the dynamic-κ modulation is interpretive overhead we do not claim as an accuracy lever — retained only because freezing it does not improve accuracy either (consistent with the §4.5 mechanism retraction). Unlike the 40-station pilot — where fixed-κ was marginally better at 7 days — the sign is now (if negligibly) in favour of the dynamic version, which we read as within-noise.

5. Discussion

The two settings deliver a single message: the value of a physics-guided learned forecaster depends on whether the system has forecast headroom. On near-perfectly persistent reservoir outlets, damped persistence is a ceiling and the honest result is parity-minus on point accuracy with a calibration-only contribution. On a diverse large sample, the same model beats the physics baselines, transfers to unseen basins, and delivers near-nominally-calibrated warnings — while being on par with, not superior to, a strong gradient-boosting learner. This argues against single-site "state-of-the-art" claims and for large-sample, transfer-tested evaluation as the standard for this problem.

Limitations. The large-sample model omits reservoir level (gage height is unavailable at most USGS temperature gages, so the rating-curve physics line is inactive at scale) and uses Daymet incident solar radiation as the radiative channel `DH` (a physical replacement of unknown scale relative to the original 3-station `DH`); wind speed is included from gridMET in the 7-variable configuration. The median margin over damped persistence at 3–7 days is small (the win-rate carries the result), and a strong gradient-boosting baseline (LightGBM) is at least as accurate as ThermoRoute: significantly better at 1 day and statistically indistinguishable at 3–7 days once the seed budget is matched (per-seed protocol, §4.2). We therefore claim a robust improvement over the physics baselines and calibrated, transferable uncertainty — not a new state of the art over all learners. We report five seeds. The dynamic-κ thermal-memory modulation does not generalise: the flow-dependence seen on three stations vanishes on the large sample (κ rises with flow at 0 % of stations), and freezing κ's modulators does not worsen RMSE, so we retract the mechanism claim and keep only the router's interpretable, horizon-dependent driver shares.

Input uncertainty. Our forcings are gridded reanalysis (Daymet, gridMET), which we treat as exact inputs — the very assumption that BATEA [G2, G3] warns against for rainfall–runoff, where forcing error can dominate and, if ignored, biases the calibrated parameters. Two points make this less acute here than in the rainfall–runoff setting BATEA addresses. First, water temperature is far smoother and more strongly autocorrelated than discharge, so short-horizon skill is carried by the persistent state (`WTEMP` history) rather than by the noisy meteorological forcing, limiting the leverage of input error. Second, we do not need the input-error model that BATEA requires and that it concedes is "poorly understood": the conformal layer calibrates the predictive intervals empirically on held-out years, absorbing residual forcing error into the coverage rather than attributing it. The cost, in BATEA's terms, is that we cannot separate forcing error from model error — a deliberate trade of process attribution for distribution-free coverage at scale. A fully operational system would additionally replace our reanalysis forcings with archived numerical weather-prediction forecasts, whose own error would degrade multi-day skill and could be handled either by a BATEA-style latent input model or by re-calibrating the conformal intervals on forecast-driven residuals; we leave this operational-forcing track to future work.

Equifinality. The near-identical validation loss across our five seeds (1.343–1.348) is a learned-model echo of the parameter equifinality that motivated GLUE [G1]: many weight configurations simulate the calibration data almost equally well. Where GLUE embraces this and propagates it into the prediction limits, our bounded physics prior instead reduces it — constraining the residual so the admissible solutions cluster near damped persistence — which is philosophically closer to BATEA's add-structure stance than to GLUE's accept-equifinality one, and is consistent with the low across-seed spread we observe.

6. Conclusions

Under a strict historical-information protocol, ThermoRoute matches per-station damped persistence where the system is near-deterministic (a reservoir cascade) and beats the physics baselines where forecast headroom exists (a 120-station large sample), transferring to unseen basins and providing near-nominally-calibrated warnings. We deliberately report two negative results — no point-accuracy gain on the cascade, and no generalisable flow-dependent thermal memory — to keep the claims honest. The contribution is a calibrated, transferable, interpretable forecaster whose advantage over the physics baselines is established on a large, diverse sample rather than a single site, with the explicit caveat that a strong gradient-boosting learner is at least as accurate across leads — better at 1 day and statistically on par at 3–7 days — so the claim is calibrated, transferable skill over the physics baselines, not a new state of the art over all learners.

Data availability

This study uses two datasets.

(1) Three-station reservoir cascade (case study). Daily records (2006-01-01 to 2020-12-31; 5 479 days per station) of water temperature, discharge, reservoir level, air temperature, precipitation, wind speed, mean relative humidity and a radiative index (`DH`) at three cascade stations (b1, s2, p3). These were provided for this study; the `DH` field's data-dictionary definition is unconfirmed and we make no `DH`-specific physical claim. Two sentinel missing-codes (`WDSP=999.9`, `PRCP=99.99`; 0.1–0.3 % of records) are masked to NaN and imputed within the training fold.

(2) 120-station USGS large sample (main analysis). Assembled programmatically from three open sources, in the same schema, by `scripts/data_usgs/build_usgs_stations.py`:

study variable	source	access	notes
`WTEMP`, `FLOW`, `WLEVEL`	USGS NWIS daily values	`dataRetrieval`/`dataretrieval-python` (U.S. Geological Survey); public domain	parameter codes 00010 / 00060 / 00065. Gage height (`WLEVEL`) is unavailable at most temperature gages (≈0 % coverage), so the large-sample model omits it; consequently the stage–discharge (rating-curve) physics is inactive at scale.
`TEMP`, `PRCP`, `DH`, `RHMEAN`	Daymet v4 (1 km gridded)	ORNL DAAC single-pixel API; open (CC0-equivalent for U.S. government / DAAC terms)	`TEMP` = mean of tmax/tmin; `DH` = incident solar radiation (`srad`, a physical radiative index replacing the original ambiguous `DH`, on a different scale); `RHMEAN` derived from vapour pressure via the Tetens saturation relation.
`WDSP`	gridMET	Climatology Lab / Northwest Knowledge Network THREDDS NetCDF-Subset Service; open	daily mean wind speed at the station coordinate.

120 stream gages across 35 U.S. states and 16 HUC2 regions (≥55 % water-temperature and ≥70 % flow coverage over 2006–2020, plus a ≥80 % blind-test-window coverage gate — §2) span free-flowing and dam-regulated rivers; water-temperature coverage ranges 0.56–1.00 (median ≈0.88). The assembled main panel is `data_usgs/panel_usgs_100.parquet` (120 stations, seven model variables plus WLEVEL; the filename is historical and predates the 120-station rebuild). Station metadata with USGS site numbers and coordinates is in `data_usgs/stations_meta_120v2.csv`, and every probed-but-rejected candidate is recorded in `data_usgs/rejected_sites_120v2.csv` (`outputs/reports/usgs_acquisition.md` documents the audit trail). The legacy 40-station pilot panels (`data_usgs/panel_usgs.parquet`, `data_usgs/panel_usgs_wind.parquet`) are retained only for the §S1 pilot comparison. All panels are included so the experiments reproduce without re-downloading; the raw per-site downloads can be regenerated by re-running the acquisition script. All three sources are public domain or open-licensed.

Code availability and reproducibility

All code, the fixed train/validation/calibration/blind-test boundaries, unit tests (leakage, splits, metrics, conformal, cross-model sample consistency), and one-command reproduction (`scripts/run_all.sh`) are in this repository. Per-day predictions, model weights and logs are regenerable by the staged scripts (`scripts/01–14`); `scripts/14_manifest.py` writes a sha256 manifest of every artifact the manuscript's numbers depend on (`outputs/manifest.json`), and `--check` verifies the recorded hashes so any drift between the manuscript and the artifacts is machine-detectable. Continuous integration runs the test suite under both the version-locked environment that produced the published numbers (`requirements-lock.txt`) and loose forward-compatibility floors. An adversarial internal review (multiple independent expert lenses) checked every headline claim against the result tables; all negative results and ablations are retained, and a finding-by-finding disposition is provided in `outputs/reports/review_response.md`.

Archived release. For a persistent, citable record the code, configuration, the input station panels, and all derived tables/reports/figures are archived at Zenodo (https://doi.org/REPLACE_WITH_ZENODO_DOI, v1.0.0, MIT license), built with `scripts/make_release_archive.sh` and carrying the sha256 manifest; the multi-hundred-MB predictions and model checkpoints are excluded but regenerable from the archive via `scripts/run_all.sh`. The raw observations are redistributed in derived form only and are independently available from USGS NWIS, Daymet (ORNL DAAC, https://doi.org/10.3334/ORNLDAAC/2129) and gridMET; USGS site numbers for all 120 gages are in the archive. (`outputs/reports/data_availability.md` holds the submission-ready Open Research statement; the DOI is minted at camera-ready.)

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